

## GAYLORD 9 SSW, MI, USA

WMO: 725410

Lat: 44.9080N

Lon: 84.7203W

Elev: 1461

StdP: 13.94

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 54854

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.1       | 0.6 | -11.5                         | 3.1 | -3.7 | -6.3 | 4.1 | 0.6  | 12.8                  | 24.4 | 11.6 | 22.8 | 3.5                   | N/A  | 0.491 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.0                   | 84.3            | 68.3 | 81.7 | 66.7 | 78.9 | 65.2 | 71.5                | 80.2 | 69.8 | 77.9 | 68.2 | 75.4 | 5.1                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 68.9                            | 112.0 | 76.0 | 67.2 | 105.7 | 73.8 | 65.7 | 100.2 | 72.0 | 36.3          | 80.2 | 34.7 | 77.8 | 33.3 | 75.2 | 75.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.9              | 9.7  | 8.6 | -11.4                      | 88.9 | 6.2                | 2.2 | -15.8                                              | 90.4 | -19.5      | 91.7 | -23.0      | 92.9 | -27.5      | 94.5 |    |    |
|                   |      |     | -12.9                      | 74.5 | 6.7                | 1.5 | -17.7                                              | 75.5 | -21.6      | 76.4 | -25.4      | 77.2 | -30.2      | 78.2 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 43.2  | 17.8  | 19.0  | 28.4  | 40.6  | 54.3  | 61.9  | 66.5  | 65.3  | 58.8  | 46.7  | 34.1  | 23.9 | (6)  |
| (7)  |                                                                   | DBStd       | 19.68 | 10.45 | 10.46 | 11.89 | 10.11 | 9.57  | 6.82  | 5.75  | 5.78  | 7.88  | 9.05  | 9.79  | 8.91 | (7)  |
| (8)  |                                                                   | HDD50       | 4418  | 999   | 869   | 682   | 315   | 64    | 4     | 0     | 0     | 13    | 179   | 485   | 808  | (8)  |
| (9)  |                                                                   | HDD65       | 8221  | 1464  | 1289  | 1137  | 732   | 354   | 136   | 52    | 67    | 218   | 573   | 926   | 1273 | (9)  |
| (10) |                                                                   | CDD50       | 1951  | 0     | 0     | 11    | 33    | 197   | 361   | 512   | 476   | 276   | 76    | 9     | 0    | (10) |
| (11) |                                                                   | CDD65       | 279   | 0     | 0     | 1     | 1     | 22    | 43    | 99    | 78    | 31    | 4     | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 2070  | 0     | 0     | 9     | 10    | 227   | 335   | 746   | 530   | 188   | 25    | 0     | 0    | (12) |
| (13) |                                                                   | CDH80       | 405   | 0     | 0     | 1     | 0     | 37    | 66    | 153   | 106   | 40    | 2     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.2   | 4.7   | 4.8   | 5.0   | 5.3   | 4.7   | 3.6   | 3.0   | 3.0   | 3.2   | 4.0   | 4.5   | 4.6  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 33.0  | 2.2   | 1.6   | 1.7   | 2.7   | 3.2   | 3.5   | 3.2   | 3.2   | 3.3   | 3.4   | 2.8   | 2.1  | (15) |
| (16) |                                                                   | PrecMax     | 39.2  | 3.6   | 2.7   | 4.5   | 4.0   | 5.3   | 7.0   | 7.7   | 5.5   | 5.5   | 8.7   | 5.4   | 4.4  | (16) |
| (17) |                                                                   | PrecMin     | 27.8  | 0.7   | 0.5   | 0.4   | 0.8   | 0.8   | 1.1   | 1.3   | 1.7   | 0.9   | 1.1   | 1.1   | 0.7  | (17) |
| (18) |                                                                   | PrecStd     | 3.5   | 0.9   | 0.6   | 0.9   | 0.9   | 1.1   | 1.5   | 1.4   | 1.1   | 1.1   | 1.7   | 1.3   | 0.9  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 47.0  | 49.9  | 73.3  | 75.3  | 84.0  | 85.7  | 87.8  | 86.7  | 85.0  | 77.9  | 65.1  | 52.6 | (19) |
| (20) |                                                                   | MCWB        | 44.8  | 45.5  | 60.0  | 55.2  | 66.4  | 70.3  | 70.5  | 68.3  | 69.2  | 59.7  | 54.0  | 49.7  | (20) |      |
| (21) |                                                                   | 2%          | DB    | 40.7  | 42.0  | 58.8  | 68.8  | 80.3  | 81.7  | 84.1  | 83.3  | 79.3  | 71.1  | 58.0  | 43.4 | (21) |
| (22) |                                                                   | MCWB        | 38.1  | 37.9  | 52.1  | 54.2  | 62.1  | 65.9  | 69.1  | 66.8  | 66.4  | 60.6  | 50.5  | 40.8  | (22) |      |
| (23) |                                                                   | 5%          | DB    | 35.6  | 36.9  | 50.3  | 62.2  | 76.0  | 77.8  | 81.1  | 79.7  | 74.9  | 65.9  | 52.6  | 39.0 | (23) |
| (24) |                                                                   | MCWB        | 33.2  | 33.9  | 43.6  | 49.7  | 60.5  | 64.4  | 67.7  | 65.7  | 64.5  | 59.0  | 46.9  | 36.9  | (24) |      |
| (25) |                                                                   | 10%         | DB    | 32.3  | 32.5  | 44.2  | 57.0  | 71.2  | 74.6  | 78.2  | 76.7  | 71.5  | 60.6  | 47.8  | 35.5 | (25) |
| (26) |                                                                   | MCWB        | 31.0  | 30.1  | 38.4  | 46.5  | 58.0  | 61.8  | 66.2  | 64.3  | 63.7  | 55.2  | 43.7  | 33.8  | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 45.0  | 46.9  | 59.8  | 60.2  | 68.7  | 73.1  | 74.8  | 71.3  | 72.3  | 66.1  | 55.9  | 52.0 | (27) |
| (28) |                                                                   | MDCB        | 46.2  | 48.1  | 71.8  | 66.6  | 79.3  | 82.4  | 84.4  | 79.4  | 82.0  | 69.3  | 60.6  | 53.6  | (28) |      |
| (29) |                                                                   | 2%          | WB    | 37.6  | 39.0  | 54.3  | 54.7  | 66.1  | 69.1  | 72.5  | 69.7  | 69.4  | 62.6  | 50.9  | 42.3 | (29) |
| (30) |                                                                   | MDCB        | 40.4  | 42.1  | 60.1  | 62.0  | 75.3  | 77.8  | 80.8  | 78.8  | 75.5  | 66.7  | 56.5  | 43.3  | (30) |      |
| (31) |                                                                   | 5%          | WB    | 33.3  | 34.8  | 44.5  | 50.0  | 63.5  | 66.8  | 70.6  | 68.2  | 67.4  | 58.7  | 47.3  | 38.0 | (31) |
| (32) |                                                                   | MDCB        | 34.8  | 37.6  | 49.6  | 57.4  | 72.3  | 73.6  | 78.2  | 75.9  | 72.4  | 64.4  | 51.8  | 39.6  | (32) |      |
| (33) |                                                                   | 10%         | WB    | 31.3  | 31.0  | 38.7  | 45.7  | 60.4  | 64.7  | 68.6  | 66.6  | 65.4  | 55.1  | 43.6  | 34.6 | (33) |
| (34) |                                                                   | MDCB        | 32.7  | 32.9  | 43.3  | 53.4  | 68.4  | 71.8  | 75.9  | 73.3  | 70.8  | 59.6  | 47.0  | 36.3  | (34) |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 11.1  | 13.1  | 15.7  | 17.6  | 20.7  | 19.2  | 19.0  | 18.6  | 18.3  | 14.9  | 11.8  | 9.4   | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 13.6  | 16.7  | 20.5  | 26.2  | 25.1  | 22.2  | 22.0  | 22.2  | 21.1  | 22.1  | 19.0  | 14.4 | (36) |
| (37) |                                                                   | MCWBR       | 11.1  | 13.4  | 13.1  | 15.7  | 12.8  | 11.7  | 10.4  | 9.5   | 10.1  | 13.2  | 12.9  | 12.0  | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 12.3  | 16.2  | 18.3  | 22.3  | 21.5  | 18.2  | 18.4  | 17.9  | 17.8  | 20.3  | 18.1  | 13.8 | (38) |
| (39) |                                                                   | MCWBR       | 10.9  | 14.0  | 13.1  | 15.9  | 12.1  | 11.1  | 10.4  | 9.3   | 10.3  | 14.4  | 13.9  | 12.7  | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.277 | 0.285 | 0.305 | 0.336 | 0.364 | 0.382 | 0.380 | 0.371 | 0.358 | 0.324 | 0.293 | 0.267 | (40) |      |
| (41) |                                                                   | taud        | 2.325 | 2.290 | 2.413 | 2.385 | 2.366 | 2.333 | 2.357 | 2.376 | 2.412 | 2.507 | 2.527 | 2.430 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 266   | 286   | 296   | 293   | 287   | 281   | 280   | 279   | 273   | 270   | 263   | 261   | (42) |      |
| (43) |                                                                   | Edh at Noon | 25    | 31    | 32    | 36    | 38    | 39    | 38    | 36    | 32    | 25    | 21    | 20    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 391   | 684   | 1130  | 1447  | 1748  | 1913  | 1931  | 1641  | 1281  | 759   | 421   | 295   | (44) |      |
| (45) |                                                                   | RadStd      | 38    | 74    | 127   | 140   | 156   | 130   | 85    | 99    | 115   | 89    | 59    | 30    | (45) |      |

## Historical Trends

|                         |     | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|-------------------------|-----|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                         |     |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)                     | (b) | (c)   | (d)     | (e)    | (f)   | (g)     | (h)   | (i) | (j)         | (k)   | (l)   | (m)   |
| Station Only            |     |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| Regional (48 neighbors) |     |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page